

torch.nn.functional.avg_pool3d#

https://pytorch.org/docs/stable/generated/torch.nn.functional.avg_pool3d.html

Description:

Applies 3D average-pooling operation in $k_T \times k_H \times k_W$ $\times k_H \times k_W$ regions by step size $s_T \times s_H \times s_W$ $\times s_H \times s_W$ steps. The number of output features is equal to $\lfloor \frac{\text{input planes}}{s_T} \rfloor \lfloor \frac{\text{input planes}}{s_H} \rfloor \lfloor \frac{\text{input planes}}{s_W} \rfloor$. See AvgPool3d for details and output shape. input – input tensor (minibatch, in_channels, iT×iH, iW) (minibatch, in_channels, iT×iH, iW) kernel_size – size of the pooling region. Can be a single number or a tuple (kT, kH, kW) stride – stride of the pooling operation. Can be a single number or a tuple (sT, sH, sW). Default: kernel_size

Function Signature

```
torch.nn.functional.avg_pool3d(input, kernel_size,
stride=None, padding=0, ceil_mode=False,
count_include_pad=True, divisor_override=None) → Tensor#
```

Parameters

Detailed Documentation

Documentation

Applies 3D average-pooling operation in $k_T \times k_H \times k_W$ $\times$ k_H $\times$ k_W $\times$ $k_H \times k_W$ regions by step size $s_T \times s_H \times s_W$ $\times$ s_H $\times$ s_W $\times$ $s_H \times s_W$ steps. The number of output features is equal to $\lfloor \frac{\text{input planes}}{s_T} \rfloor \lfloor \frac{\text{input planes}}{s_H} \rfloor \lfloor \frac{\text{input planes}}{s_W} \rfloor$.

See AvgPool3d for details and output shape.

`input` – input tensor (`minibatch, in_channels, iT × iH, iW`) (`minibatch`, `in_channels`, `iT × iH, iW`)

`kernel_size` – size of the pooling region. Can be a single number or a tuple (`kT, kH, kW`)

`stride` – stride of the pooling operation. Can be a single number or a tuple (`sT, sH, sW`).

Default: `kernel_size`

`padding` – implicit zero paddings on both sides of the input. Can be a single number or a tuple (`padT, padH, padW`). Should be at most half of effective kernel size, that is $\frac{((\text{kernelSize} - 1) * \text{dilation} + 1)}{2}$. Default: 0

`ceil_mode` – when True, will use ceil instead of floor in the formula to compute the output shape

`count_include_pad` – when True, will include the zero-padding in the averaging calculation

`divisor_override` – if specified, it will be used as divisor, otherwise size of the pooling

region will be used. Default: None